

Scalability Implementation Summary

1. Frontend Scalability

This requirement has already been addressed through the use of Vercel. Since the frontend is deployed on Vercel's serverless infrastructure, it automatically scales to handle increasing traffic without requiring manual configuration. As a result, frontend scalability is effectively managed by the platform.

2. Backend Scalability (API Layer)

The backend is also deployed on Vercel, which provides serverless scaling for API requests. This allows multiple requests to be handled in parallel without relying on a single server instance. While traditional horizontal scaling and load balancing are mentioned in the plan, Vercel abstracts these concerns by automatically scaling API endpoints.

3. Database Scalability (Supabase)

Based on earlier discussion, advanced database optimizations such as read replicas and connection pooling were not necessary for the current scale of the project. Supabase already provides a managed and optimized PostgreSQL database, and the current workload does not require further database-level scaling optimizations.

4. Workflow Scalability (n8n Automation)

Workflow scalability was improved by modifying the n8n pipeline to process multiple emails in a loop rather than handling a single email per execution. Additionally, multiple workflows were tested and refined to ensure reliable concurrent processing. The system was also updated to use Groq instead of Gemini, improving performance and throughput for AI-related tasks.

5. AI API Scaling

This area required minimal changes, as the system was already performing efficiently. Switching from Gemini to Groq helped improve speed and reduce latency, and the available token limits were sufficient for current usage. As a result, additional optimizations such as caching or batching were not necessary at this stage.

6. Asynchronous Processing

This requirement was largely satisfied through the use of n8n workflows, which handle email processing and AI analysis outside of the main user request flow. This ensures that heavy operations do not block user-facing interactions.

7. Monitoring and Observability

This has not yet been implemented. While it was researched, tools for monitoring system performance, logging, and alerting have not been integrated into the current system.

8. Multi-User / Multi-Agent Scalability

The system is designed to support multiple users and concurrent email processing. While not

fully stress-tested, the current architecture using serverless deployment and concurrent workflows should support multiple users interacting with the system simultaneously.